

Notes: Barberis, Shleifer & Wurgler (JFE, 2005)

Comovement

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1 Big picture

- **Question.** Why do some stocks move together? Is return comovement mainly about **common news about fundamentals**, or can it be created by **investor demand, preferred clienteles, or trading frictions**?
- **Setting.** The paper studies **changes in S&P 500 membership**. If a stock enters a highly salient index category and then starts moving more with that category, this is hard to reconcile with a purely fundamentals-based view of comovement.
- **Core empirical challenge.** A stock added to the S&P 500 is often large, liquid, and increasingly important already. So the paper must separate a change in **return comovement caused by index membership** from a change caused by **size, industry, liquidity, or evolving fundamentals**.
- **Main conceptual contribution.** The paper places a very clean empirical design next to a simple theoretical contrast:
 - under the **fundamentals view**, inclusion should not change return covariances if cash-flow covariance is unchanged;
 - under **category, habitat, or information-diffusion** views, inclusion should make the stock comove more with the S&P and less with stocks outside the S&P.
- **Main theories.**
 - **Category view:** investors think in categories such as “S&P 500 stocks” and move money across categories.
 - **Habitat view:** some investors trade only a subset of securities, so shocks to those investors affect that subset together.
 - **Information-diffusion view:** some stocks incorporate aggregate news faster than others; co-movement partly reflects similar adjustment speed.
- **Main empirical design.** For each inclusion or clean deletion, the paper estimates **pre-event** and **post-event** betas using univariate and bivariate regressions at daily, weekly, and monthly frequencies, and then checks the results with **calendar-time portfolios, matched firms, turnover-based non-trading tests, and Dimson lead-lag regressions**.
- **Main baseline result.** For **455 inclusions** during 1976–2000, daily event-time regressions show:
 - univariate S&P beta rises by about **0.151**;
 - the univariate R^2 rises by about **0.049**;
 - in bivariate regressions, S&P beta rises by about **0.326**;
 - beta on non-S&P stocks falls by about **0.319**.
- **Main time-variation result.** The effects are **stronger in the later sample**. For additions, the daily univariate beta increase is only **0.067** in 1976–1987, but **0.214** in 1988–2000. This

is exactly what one would expect if the S&P became a more important trading category over time.

- **Deletion result.** Clean deletions are much harder to find, but in the bivariate daily regressions the sign flips as predicted: S&P beta falls by about **0.511** and non-S&P beta rises by about **0.550**.
- **Main credibility result.** The paper shows that the basic result survives:
 - calendar-time estimation,
 - matching on **industry** and **recent size growth**,
 - tests against **non-synchronous trading**, and
 - lead-lag adjustments that isolate **information-diffusion** effects.
- **Mechanism result.** Information diffusion explains **some** of the daily effect, especially in the bivariate regressions, but not all of it. After controlling for lead-lag effects, there is still a substantial residual change in comovement, which points toward **category/habitat demand effects**.
- **Economic interpretation.** Index membership affects not only **price levels** but also the **correlation structure of returns**. A stock starts moving with the investor clientele and trading habitat attached to the index it joins.
- **Main takeaway.** Comovement is not just about common fundamentals. Market frictions, investor categorization, and preferred habitats can reshape betas in economically important ways.

2 Data and measurement (institutional context and key objects)

2.1 Why the S&P 500 is the right laboratory

- The S&P 500 is a natural **category** and **habitat**: index funds, futures, and options make it a heavily used trading benchmark.
- S&P stocks are also unusually liquid and institutionally followed, which makes them plausible “fast-information” stocks under the information-diffusion view.
- There is clear turnover in membership, roughly **about 30 changes per year**.
- Standard & Poor’s states that the goal of inclusion is to keep the index **representative of the U.S. economy**, not to announce private information about future cash flows.

Interpretation. The S&P 500 gives the paper exactly what it needs: a salient group, observable reclassification events, and a plausible argument that index entry should not mechanically change a firm’s fundamental covariance with the market.

2.2 Sample construction

- Inclusion sample period: **September 22, 1976 to December 31, 2000.**
- Deletion sample period: **January 1, 1979 to December 31, 2000.**
- Raw sample: **590 inclusion events** and **565 deletion events.**
- Inclusions are excluded if the new firm is a spin-off, a restructuring of a firm already in the index, involved in a merger/takeover around the event, or too near the sample end to leave enough post-event data.
- Final inclusion sample:
 - **455** events for daily and weekly tests;
 - **324** events for monthly tests.
- Deletions are much dirtier because many are tied to mergers, acquisitions, or bankruptcy. After removing these cases, the final deletion sample contains:
 - **76** events for daily and weekly tests;
 - **45** events for monthly tests.

Interpretation. The paper is much more persuasive on additions than deletions because clean deletions are scarce. The authors knowingly sacrifice sample size to keep the design credible.

2.3 Event timing and estimation windows

- Before **October 1989**, inclusions and deletions were effective **on the announcement day.**
- After October 1989, changes were typically announced about **one week before implementation.**
- To avoid ambiguity about whether a stock should count as “in” the index between announcement and implementation, the paper excludes the **announcement month** and the **implementation month** from estimation windows.
- For daily and weekly data, pre-event regressions use the **12 months** ending the month before announcement; post-event regressions use the **12 months** starting the month after implementation.
- For monthly data, the paper uses **36-month** pre- and post-event windows.
- Returns from **October 1987** are excluded because the crash would otherwise dominate the results.

Interpretation. The timing choices are careful. The authors want clean pre/post regimes, not a noisy transition period contaminated by announcement-day price pressure or institutional timing conventions.

2.4 Main return variables

For event stock j , the left-hand-side variable is simply its return:

$$R_{j,t}. \quad (1)$$

The paper’s main right-hand-side variables are:

- $R_{SP500,t}$: the return on the S&P 500, taken from the CRSP Index on the S&P Universe file;
- $R_{nonSP500,t}$: the capitalization-weighted return on NYSE/AMEX/Nasdaq stocks outside the S&P 500.

The non-S&P return is inferred from the identity

$$R_t^{VWCRSP} = \left(\frac{CAP_{CRSP,t-1} - CAP_{SP500,t-1}}{CAP_{CRSP,t-1}} \right) R_{nonSP500,t} + \left(\frac{CAP_{SP500,t-1}}{CAP_{CRSP,t-1}} \right) R_{SP500,t}. \quad (2)$$

- The **mechanical contribution of the event stock** is removed from the right-hand-side index return as appropriate.
- Regressions are estimated at **daily**, **weekly**, and **monthly** frequencies.

Interpretation. The paper is not just regressing on “the market.” It explicitly separates comovement with **S&P stocks** from comovement with **non-S&P stocks**, which is what makes the bivariate test so sharp.

2.5 Main outcome measures

The paper focuses on four central objects.

- **Univariate S&P beta** b_j : how much the stock moves with the S&P.
- **Univariate R^2** : how much of the stock’s return variation is explained by the S&P.
- **Bivariate S&P beta** $b_{j,SP500}$: the stock’s loading on the S&P after controlling for non-S&P stocks.
- **Bivariate non-S&P beta** $b_{j,nonSP500}$: the stock’s loading on the non-S&P market after controlling for the S&P.

Interpretation. The univariate tests ask whether the stock moves more with the index; the bivariate tests ask whether it moves more with the index *rather than* with the rest of the market.

2.6 Matched-firm variables

- For each event stock, the authors search for a non-event stock matched on:
 - **industry**, and

- **market capitalization decile** both at the event date and **12 months earlier** (or **36 months earlier** for monthly tests).
- For deletion events, the matching stock is another S&P member with similar industry and recent size path but that is **not** deleted.

Interpretation. This is the paper’s main defense against the claim that event firms simply happen to be on a size or industry trajectory that would have changed their comovement anyway.

2.7 Turnover and trading-activity measures

- To test whether daily beta changes are just a non-trading artifact, the paper computes each stock’s average monthly **turnover**:

$$Turnover = \frac{\text{Share volume}}{\text{Shares outstanding}}. \quad (3)$$

- It then splits additions into stocks whose turnover **falls** after inclusion and stocks whose turnover **rises** after inclusion.

Interpretation. If beta increases only because stocks trade more often after inclusion, then stocks with lower post-event turnover should not show the same positive beta shifts. This becomes a direct test of the non-trading story.

2.8 A few descriptive facts worth remembering

- Additions are much cleaner than deletions because deletions often happen when the firm itself is changing dramatically.
- The later sample is especially informative because S&P-linked investment products became much more important over time.
- The paper is not claiming that inclusion never coincides with information about the *level* of future cash flows; rather, it argues that inclusion should not obviously change the *covariance structure* of those cash flows.

Interpretation. The entire design hinges on a narrower claim than “inclusion contains no information at all.” The crucial claim is: inclusion should not by itself make the firm’s fundamentals more correlated with S&P fundamentals.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing theories of comovement

The paper contrasts four broad ideas.

- **Fundamentals-based view.** Assets comove because news about their discounted cash flows comoves.
- **Category view.** Investors allocate money at the level of categories such as “S&P 500 stocks,” creating common price pressure inside the category.
- **Habitat view.** Particular investor clienteles trade only a subset of securities, so shocks to those clienteles generate common movement in that subset.
- **Information-diffusion view.** Some stocks incorporate aggregate news faster than others, so stocks with similar information speed move together.

Interpretation. The first view says comovement is about common *fundamentals*. The other three say it can arise from common *investor demand or trading frictions*.

3.2 Dividend and return setup

Each risky asset pays a terminal dividend

$$D_{i,T} = D_{i,0} + e_{i,1} + \cdots + e_{i,T}, \quad (4)$$

where the $e_{i,t}$ terms are cash-flow shocks.

The paper writes the asset’s one-period return in reduced form as the change in price:

$$\Delta P_{i,t} = P_{i,t} - P_{i,t-1}. \quad (5)$$

Interpretation. The theory is deliberately reduced form. The goal is not full structural estimation, but clear predictions about what should happen to betas when a stock is reclassified into a salient group.

3.3 Category and habitat reduced forms

If assets are grouped into categories or habitats X and Y , returns can be written as

$$\Delta P_{i,t} = e_{i,t} + \Delta u_{X,t}, \quad i \in X, \quad (6)$$

$$\Delta P_{j,t} = e_{j,t} + \Delta u_{Y,t}, \quad j \in Y, \quad (7)$$

where $u_{X,t}$ and $u_{Y,t}$ capture sentiment, fund flows, or investor-demand shocks specific to the two groups.

Interpretation. Even if two stocks have unrelated cash flows, they can move together if investors trade them together as part of the same category or habitat.

3.4 Information-diffusion reduced form

Under the information-diffusion view, one group adjusts quickly while another adjusts with delay:

$$\Delta P_{i,t} = e_{i,t}, \quad i \in X, \quad (8)$$

$$\Delta P_{j,t} = m e_{j,t} + (1 - m) e_{j,t-1}, \quad j \in Y. \quad (9)$$

Interpretation. Stocks in X impound news immediately; stocks in Y partially catch up later. This delayed adjustment itself creates comovement patterns that need not come from fundamentals.

3.5 Prediction 1: univariate reclassification test

The paper's first test is a univariate regression of event-stock returns on the return of the destination group:

$$\Delta P_{j,t} = a_j + b_j \Delta P_{X,t} + v_{j,t}, \quad (10)$$

where

$$\Delta P_{X,t} = \frac{1}{n} \sum_{\ell \in X} \Delta P_{\ell,t}. \quad (11)$$

The theoretical prediction is:

- if stock j moves from group Y into group X , then its **beta on X** and its R^2 should rise;
- if comovement is purely fundamentals-based and cash-flow covariance is unchanged, these objects should **not** change.

Interpretation. This is the direct translation of the paper's theory into data: reclassification should alter *return comovement* even if it does not alter *fundamental covariance*.

3.6 Prediction 2: bivariate reclassification test

The paper's main test is the bivariate regression

$$\Delta P_{j,t} = a_j + b_{j,X} \Delta P_{X,t} + b_{j,Y} \Delta P_{Y,t} + v_{j,t}. \quad (12)$$

In the S&P application this becomes the regression of stock j on **S&P returns** and **non-S&P returns**.

The theoretical prediction is:

- after reclassification into X , $b_{j,X}$ **rises**;
- $b_{j,Y}$ **falls**;
- under the fundamentals view, neither coefficient should shift systematically.

Interpretation. This regression is more powerful because it asks not merely “does the stock move more with the S&P?” but “does it move more with the S&P *relative to the rest of the market*?”

3.7 Event-time empirical regressions

For each event stock j , the paper estimates the univariate regression

$$R_{j,t} = a_j + b_j R_{SP500,t} + v_{j,t}, \quad (13)$$

and records the changes

$$\Delta b_j = b_j^{post} - b_j^{pre}, \quad \Delta R_j^2 = (R_j^2)^{post} - (R_j^2)^{pre}. \quad (14)$$

It also estimates the bivariate regression

$$R_{j,t} = a_j + b_{j,SP500} R_{SP500,t} + b_{j,nonSP500} R_{nonSP500,t} + v_{j,t}, \quad (15)$$

and records

$$\Delta b_{j,SP500}, \quad \Delta b_{j,nonSP500}. \quad (16)$$

Interpretation. The entire paper turns on whether these average pre/post changes are systematically positive for S&P exposure and negative for non-S&P exposure.

3.8 Calendar-time portfolio tests

As a robustness check, the paper constructs a **pre-event portfolio** of stocks soon to be added and a **post-event portfolio** of stocks recently added. It then estimates

$$R_{pre,t} = a_{pre} + b_{pre} R_{SP500,t} + v_{pre,t}, \quad (17)$$

$$R_{post,t} = a_{post} + b_{post} R_{SP500,t} + v_{post,t}, \quad (18)$$

and the bivariate analogues

$$R_{pre,t} = a_{pre} + b_{pre,SP500} R_{SP500,t} + b_{pre,nonSP500} R_{nonSP500,t} + v_{pre,t}, \quad (19)$$

$$R_{post,t} = a_{post} + b_{post,SP500} R_{SP500,t} + b_{post,nonSP500} R_{nonSP500,t} + v_{post,t}. \quad (20)$$

Interpretation. The calendar-time approach is useful because event windows overlap. It gives a different way to address cross-event dependence.

3.9 Matching-firm design

For each event stock and matching stock, the paper computes the same pre/post beta changes, then looks at the **difference in those changes**:

$$\Delta \Delta b = \Delta b^{event} - \Delta b^{match}, \quad (21)$$

with analogous objects for ΔR^2 , Δb_{SP500} , and $\Delta b_{nonSP500}$. **Interpretation.** This is the closest thing in the paper to a difference-in-differences test. It asks whether event stocks shift more than

otherwise similar stocks that were not reclassified.

3.10 Non-trading test

The daily-frequency non-trading hypothesis predicts that beta should rise mainly when the stock's trading frequency rises after inclusion. The paper therefore splits the sample by whether turnover **increases** or **decreases** after inclusion. **Interpretation.** If beta rises even when turnover falls, then non-synchronous trading cannot be the main explanation.

3.11 Lead-lag regressions for information diffusion

To isolate information-diffusion effects, the paper estimates Dimson-style regressions with five leads and lags:

$$R_{j,t} = a_j + \sum_{s=-5}^5 b_j^{(s)} R_{SP500,t+s} + v_{j,t}, \quad (22)$$

and

$$R_{j,t} = a_j + \sum_{s=-5}^5 b_{j,SP500}^{(s)} R_{SP500,t+s} + \sum_{s=-5}^5 b_{j,nonSP500}^{(s)} R_{nonSP500,t+s} + v_{j,t}. \quad (23)$$

The weighted sums of these coefficients are interpreted as “no-delay” betas. **Interpretation.** If the raw daily beta shifts shrink sharply after adding leads and lags, then part of the baseline result comes from different information-adjustment speeds. If a large residual remains, category/habitat forces are still needed.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why S&P additions are the paper's cleanest source of variation

- Reclassification into the S&P is visible and discrete.
- The index is economically important enough that category/habitat effects are plausible.
- Standard & Poor's does not claim to be revealing new information about the firm's risk covariance; it claims to be keeping the index representative.
- This makes the change in **return comovement** more interpretable as a market-structure effect rather than a fundamentals effect.

Bottom line. The identification comes from a before/after shift in group membership where the author's null is that group membership should not by itself change fundamental covariance.

4.2 What exactly identifies the main estimates

- Each event stock is compared to itself before and after inclusion or deletion.
- The paper removes the event stock's own mechanical weight from the right-hand-side index return.
- The bivariate regressions explicitly hold fixed comovement with the **rest of the market**.
- Calendar-time regressions verify that the results do not depend on the specific event-time estimator.

Bottom line. The key estimate is not a cross-sectional comparison of S&P and non-S&P firms; it is the **change in the same firm's factor loadings** around reclassification.

4.3 Why a purely fundamentals-based explanation is hard to maintain

- Under the fundamentals view, inclusion should matter only if it coincides with a change in the covariance of the firm's cash flows with S&P firms' cash flows.
- The strongest facts against that explanation are:
 - the large shift in bivariate betas;
 - the stronger effects in the later sample, when index-linked trading is more important;
 - the weaker shifts in matched firms that look similar on industry and size path.

Bottom line. The paper does not prove that fundamentals never change around inclusion, but it shows that a simple fundamentals-only story struggles to explain the full pattern.

4.4 Alternative explanations and the paper's responses

1. Size or industry effects.

- Concern: additions may just be firms that are becoming larger or more representative of an increasingly dominant industry.
- Response: compare event stocks to **matched firms** from the same industry with similar recent size evolution.
- Result: event stocks still show much larger beta shifts.

2. Non-synchronous trading.

- Concern: stocks outside the S&P trade less often, which biases daily betas down before inclusion.
- Response: split the sample by whether turnover rises or falls after inclusion.
- Result: even stocks whose turnover **falls** after inclusion still show strongly positive S&P beta shifts.

3. Cross-event correlation.

- Concern: event windows overlap in calendar time, so event-specific estimates are not independent.
- Response: simulation-based standard errors and calendar-time portfolio regressions.

4. Information-diffusion effects.

- Concern: some of the daily effect may reflect faster incorporation of market news rather than category/habitat demand.
- Response: lead-lag regressions.
- Result: information diffusion matters, especially for the daily bivariate effects, but it does not explain everything.

4.5 What the paper can and cannot distinguish

- The paper can convincingly distinguish **fundamentals-only** comovement from **friction/sentiment-based** comovement.
- It can identify a role for **information diffusion** because that mechanism implies lead-lag structure in returns.
- It cannot sharply distinguish **category** from **habitat** using returns alone, because both generate similar contemporaneous beta shifts.

Bottom line. The main identification success is broad but not fully mechanism-specific: the paper identifies non-fundamental comovement, then partially decomposes it.

4.6 Limitations worth remembering

- Clean deletions are rare, so addition results are more persuasive than deletion results.
- Monthly results are weaker in the raw event-time regressions and become strongest mainly in later years.
- Bivariate regressions naturally involve collinearity between S&P and non-S&P returns, which makes coefficient interpretation less direct even though the sign pattern is still informative.
- Matching is helpful, but it cannot perfectly eliminate every omitted fundamentals story.

Bottom line. The paper is strongest as evidence that comovement is *not only* about fundamentals, not as a complete structural decomposition of all forces behind betas.

4.7 Relation to the literature

- The direct predecessor is **Vijh (1994)**, who also studies beta changes around S&P additions.

- The paper also connects to the broader index-inclusion price-effect literature, such as **Harris and Gurel (1986)**, **Shleifer (1986)**, and **Wurgler and Zhuravskaya (2002)**.
- Its contribution is to move from **price-level effects** to **comovement effects** and to propose a sharper **bivariate** test.

Bottom line. The paper helps shift the conversation from “does index demand move prices?” to “does index demand also move *betas and covariance structure*?”

5 Tables and figure readings

5.1 Table 1: core event-time evidence

Table 1

Changes in comovement of stocks added to and deleted from the S&P 500 index
Changes in the slope and the fit of regressions of returns of stocks added to and deleted from the S&P 500 index on returns of the S&P 500 index and the non-S&P 500 rest of the market. The sample includes stocks added to and deleted from the S&P 500 between 1976 and 2000 that were not involved in mergers or related events (described in the text) and that have sufficient return data on CRSP. For each event stock j , the univariate model

$$R_{j,t} = \alpha_j + \beta_j R_{SP500,t} + v_{j,t}$$

and the bivariate model

$$R_{j,t} = \alpha_j + \beta_{j,SP500} R_{SP500,t} + \beta_{j,nonSP500} R_{nonSP500,t} + v_{j,t}$$

are separately estimated for the pre- and post-event period. Returns on the S&P 500 (R_{SP500}) are from the CRSP Index on the S&P 500 Universe file. Returns on a capitalization-weighted index of the non-S&P 500 stocks ($R_{nonSP500}$) in the NYSE, AMEX, and Nasdaq are inferred from the identity

$$R_{VWCRSP,t} = \left(\frac{CAP_{CRSP,t-1} - CAP_{SP500,t-1}}{CAP_{CRSP,t-1}} \right) R_{nonSP500,t} + \left(\frac{CAP_{SP500,t-1}}{CAP_{CRSP,t-1}} \right) R_{SP500,t}$$

Total capitalization of the S&P 500 (CAP_{SP500}) is from the CRSP Index on the S&P 500 Universe file. Returns on the value-weighted CRSP NYSE, AMEX, and Nasdaq index (R_{VWCRSP}) and total capitalization (CAP_{CRSP}) are from the CRSP Stock Index file. Returns from October 1987 are excluded. The mechanical influence of the added or deleted stock is removed from the independent variables as appropriate. For the univariate regression model, we examine the mean change in slope across the event date $\Delta\beta$, and the mean change in fit ΔR^2 . For the bivariate model, we examine the mean changes in the slopes, $\Delta\beta_{SP500}$ and $\Delta\beta_{nonSP500}$. The pre-event and post-event estimation periods are $[-12, -1]$ and $[+1, +12]$ months for daily and weekly returns and $[-36, -1]$ and $[+1, +36]$ months for monthly returns. Panels A, B, and C show results for daily, weekly, and monthly returns, respectively. Standard errors are adjusted using simulations to account for cross-correlation, and are reported in parentheses.

Sample	N	Univariate		Bivariate		
		$\Delta\bar{\beta}$ (s.e.)	ΔR^2 (s.e.)	$\Delta\bar{\beta}_{SP500}$ (s.e.)	$\Delta\bar{\beta}_{nonSP500}$ (s.e.)	
<i>Panel A: daily returns</i>						
Additions	1976–2000	455	0.151*** (0.021)	0.049*** (0.005)	0.326*** (0.027)	−0.319*** (0.033)
	1976–1987	196	0.067*** (0.023)	0.038*** (0.008)	0.286*** (0.041)	−0.301*** (0.050)
	1988–2000	259	0.214*** (0.032)	0.058*** (0.007)	0.357*** (0.036)	−0.332*** (0.045)
Deletions	1979–2000	76	−0.087* (0.049)	−0.010 (0.007)	−0.511*** (0.111)	0.550*** (0.122)
<i>Panel B: weekly returns</i>						
Additions	1976–2000	455	0.110*** (0.029)	0.033*** (0.008)	0.174*** (0.053)	−0.119** (0.056)
	1976–1987	196	0.025 (0.036)	0.027** (0.012)	0.137 (0.094)	−0.125 (0.093)
	1988–2000	259	0.173*** (0.043)	0.037*** (0.010)	0.202*** (0.061)	−0.115* (0.069)

Table 1 (continued)

Sample		N	Univariate		Bivariate	
			$\Delta\bar{\beta}$ (s.e.)	ΔR^2 (s.e.)	$\Delta\bar{\beta}_{SP500}$ (s.e.)	$\Delta\bar{\beta}_{nonSP500}$ (s.e.)
Deletions	1979–2000	76	−0.129 (0.105)	−0.015 (0.010)	−0.505*** (0.161)	0.412** (0.169)
<i>Panel C: monthly returns</i>						
Additions	1976–1998	324	0.042 (0.041)	0.004 (0.014)	0.317*** (0.077)	−0.252*** (0.072)
	1976–1987	172	−0.010 (0.060)	0.006 (0.021)	0.267** (0.127)	−0.167* (0.116)
	1988–1998	152	0.101* (0.066)	0.000 (0.021)	0.375*** (0.113)	−0.348*** (0.107)
Deletions	1979–1998	45	0.006 (0.100)	0.001 (0.022)	0.303 (0.240)	−0.256 (0.252)

***, **, and * denote significant differences from zero at the 1%, 5%, and 10% levels in one-sided tests, respectively.

Read:

- For the full addition sample, daily univariate S&P beta rises by about **0.151** and the univariate R^2 rises by about **0.049**.
- The daily bivariate results are even sharper: S&P beta rises by **0.326**, while non-S&P beta falls by **0.319**.

- The daily univariate addition effect is much larger in 1988–2000 (**0.214**) than in 1976–1987 (**0.067**).
- For clean deletions, the bivariate daily coefficients flip sign as predicted: -0.511 on the S&P return and **+0.550** on the non-S&P return.
- Monthly univariate results are weak, but monthly bivariate results remain strong for additions: S&P beta rises by **0.317** and non-S&P beta falls by **0.252**.

Economic message. The paper’s main result is not just “stocks added to the S&P move more with the market.” It is “stocks added to the S&P move more with *the S&P itself* and less with non-S&P stocks.” That sign pattern is the central rejection of the fundamentals-only view.

5.2 Table 2: calendar-time robustness

Table 2

Calendar time estimates of changes in comovement of stocks added to and deleted from the S&P 500 index. Differences between the comovement characteristics of two portfolios of stocks: those about to be added to the S&P 500 and those just recently added. The sample includes stocks added to and deleted from the S&P 500 between 1976 and 2000 that were not involved in mergers or related events (described in the text) and that have sufficient return data on CRSP. A capitalization-weighted return index of non-S&P 500 stocks ($R_{nonSP500}$) in the NYSE, AMEX, and Nasdaq is inferred from the identity described in Table 1. Returns from October 1987 are excluded. In daily data, for example, for each day we form an equal-weighted portfolio of stocks that will be added to the S&P 500 within the next year and a portfolio of stocks that were added within the past year. We then run separate univariate regressions for each portfolio on the S&P 500 index,

$$R_{pre,t} = \alpha_{pre} + \beta_{pre} R_{SP500,t} + v_{pre,t}$$

and

$$R_{post,t} = \alpha_{post} + \beta_{post} R_{SP500,t} + v_{post,t}$$

denoting the difference in slope between the “post” and “pre” regressions as $\Delta\beta$. We also run separate bivariate regressions for each portfolio,

$$R_{pre,t} = \alpha_{pre} + \beta_{pre,SP500} R_{SP500,t} + \beta_{pre,nonSP500} R_{nonSP500,t} + v_{pre,t}$$

and

$$R_{post,t} = \alpha_{post} + \beta_{post,SP500} R_{SP500,t} + \beta_{post,nonSP500} R_{nonSP500,t} + v_{post,t}$$

denoting the difference in the slopes as $\Delta\beta_{SP500}$ and $\Delta\beta_{nonSP500}$, respectively. The mechanical influence of the “pre” and “post” portfolio stocks is removed, as appropriate, from the independent variables. In daily and weekly data, the “pre” portfolio includes stocks that will be added within one year and the “post” portfolio includes stocks that were added in the past year. In monthly data, these windows are extended to three years. We require at least ten stocks in each portfolio in order for that observation (day, week, or month) to be included in the regressions. Asymptotic standard errors are reported in parentheses.

Sample	T	Univariate $\Delta\beta$ (s.e.)	Bivariate		
			$\Delta\beta_{SP500}$ (s.e.)	$\Delta\beta_{nonSP500}$ (s.e.)	
<i>Panel A: daily returns</i>					
Additions	1976–2000	4147	0.123*** (0.013)	0.297*** (0.020)	−0.262*** (0.022)
	1976–1987	1873	0.116*** (0.019)	0.326*** (0.035)	−0.329*** (0.044)
	1988–2000	2274	0.129*** (0.017)	0.298*** (0.026)	−0.247*** (0.026)
Deletions	1979–2000	151	0.058 (0.107)	0.164 (0.190)	−0.080 (0.117)
<i>Panel B: weekly returns</i>					
Additions	1976–2000	856	0.045* (0.026)	0.167*** (0.046)	−0.154*** (0.043)
	1976–1987	387	0.041 (0.039)	−0.014 (0.094)	0.051 (0.099)
	1988–2000	469	0.049 (0.036)	0.219*** (0.054)	−0.210*** (0.048)

Table 2 (continued)

Sample		T	Univariate $\Delta\beta$ (s.e.)	Bivariate	
				$\Delta\beta_{SP500}$ (s.e.)	$\Delta\beta_{nonSP500}$ (s.e.)
Deletions	1979–2000	29	−0.082 (0.193)	−0.162 (0.537)	0.039 (0.288)
<i>Panel C: monthly returns</i>					
Additions	1976–1998	282	0.018 (0.042)	0.319*** (0.073)	−0.320*** (0.061)
	1976–1987	127	0.019 (0.047)	0.148 (0.114)	−0.143 (0.099)
	1988–1998	155	0.016 (0.065)	0.388*** (0.092)	−0.406*** (0.075)
Deletions	1979–1998	116	−0.123 (0.105)	−0.255* (0.166)	0.132 (0.132)

***, **, and * denote statistical significance at the 1%, 5%, and 10% levels in one-sided tests, respectively.

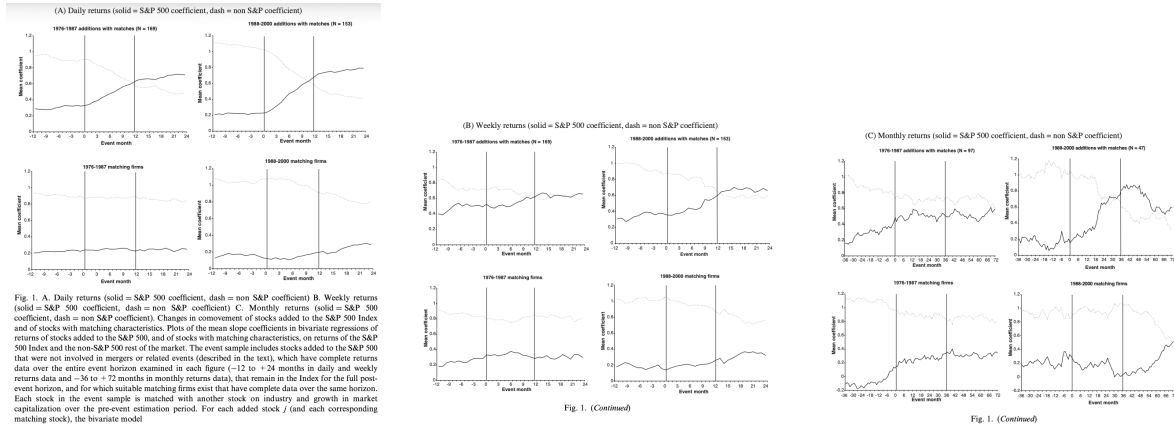
Read:

- In calendar-time daily data, univariate beta still rises by about **0.123**.
- The daily bivariate shifts are again strong: about **+0.297** for the S&P beta and -0.262 for the non-S&P beta.
- Weekly calendar-time bivariate estimates remain significant for additions.

- Monthly calendar-time bivariate estimates are also strong for additions: roughly **+0.319** on the S&P and **−0.320** on non-S&P stocks.

Economic message. The core finding is not an artifact of event-time overlap or unusual standard-error construction. It survives a calendar-time design built from portfolios of “about to enter” and “just entered” stocks.

5.3 Figure 1 and Table 3: matching-stock evidence



Read:

- The rolling-beta figure shows that event stocks experience a visible break in S&P and non-S&P loadings around inclusion, while matching firms show much smaller changes.
- In the full matched-sample daily results, the event-minus-match difference in the bivariate S&P beta change is about **0.318**; the event-minus-match difference in the non-S&P beta change is about **−0.289**.
- Even after matching, the daily univariate beta difference is still about **0.120**.
- In monthly data, the matched-sample evidence is weaker overall but stronger in the later period.

Economic message. This is the paper’s strongest response to “maybe S&P additions are just becoming more like big firms or certain industries.” Similar firms not added to the index do not show the same sharp covariance shift.

5.4 Table 4: non-trading test

Read:

- Among additions whose turnover **decreases** after inclusion, daily univariate beta still rises by about **0.083** and daily bivariate S&P beta rises by about **0.313**.
- Among additions whose turnover **increases**, the effects are even larger: daily univariate beta rises by **0.194** and daily bivariate S&P beta by **0.350**.

Table 3

Changes in comovement of stocks added to and deleted from the S&P 500 index relative to matching firms. Changes in the slope and the fit of regressions of returns of stocks added to and deleted from the S&P 500 Index on returns of the S&P 500 index and the non-S&P 500 rest of the market, relative to changes in the same parameters for matching stocks. Each stock in the event sample is matched with another stock on industry and growth in market capitalization over the pre-event estimation period. The event sample includes stocks added to and deleted from the S&P 500 between 1976 and 2000 that were not involved in mergers or related events (described in the text), that have sufficient return data on CRSP, and for which a matching stock could be found. For each event stock j , the univariate model

$$R_{j,t} = \alpha_j + \beta_j R_{SP500,t} + v_{j,t}$$

and the bivariate model

$$R_{j,t} = \alpha_j + \beta_j R_{SP500,t} + \beta_{j,nonSP500} R_{nonSP500,t} + v_{j,t}$$

are separately estimated for the pre-event and post-event period, and analogous regressions are run for each matching stock. Returns on the S&P 500 (R_{SP500}) are from the CRSP Index on the S&P 500 Universe file. Returns on a capitalization-weighted index of the non-S&P 500 stocks ($R_{nonSP500}$) in the NYSE, AMEX, and Nasdaq are inferred from the identity described in Table 1. Returns from October 1987 are excluded. The mechanical influence of the added or deleted stock is removed from the independent variables as appropriate. For the univariate regression model, we examine the mean change in slope and fit across the event date for event stocks minus the corresponding quantities for matching stocks, $\Delta\Delta\beta$ and $\Delta\Delta R^2$. For the bivariate model, we examine the mean change in slopes for event stocks minus the corresponding quantities for matching stocks, $\Delta\Delta\beta_{SP500}$ and $\Delta\Delta\beta_{nonSP500}$. The pre-event and post-event estimation periods are $[-12, -1]$ and $[+1, +12]$ months for daily and weekly returns and $[-36, -1]$ and $[+1, +36]$ months for monthly returns. Panels A, B, and C show results for daily, weekly, and monthly returns, respectively. Asymptotic standard errors are reported in parentheses.

Sample		N	Univariate		Bivariate	
			$\Delta\Delta\beta$ (s.e.)	$\Delta\Delta R^2$ (s.e.)	$\Delta\Delta\beta_{SP500}$ (s.e.)	$\Delta\Delta\beta_{nonSP500}$ (s.e.)
<i>Panel A: daily returns</i>						
Additions	1976–2000	435	0.120*** (0.022)	0.040*** (0.006)	0.318*** (0.035)	−0.289*** (0.042)
	1976–1987	189	0.109*** (0.028)	0.033*** (0.008)	0.262*** (0.051)	−0.257*** (0.065)
	1988–2000	246	0.129*** (0.032)	0.046*** (0.007)	0.361*** (0.047)	−0.313*** (0.055)
Deletions	1979–2000	36	−0.098 (0.081)	−0.012 (0.013)	−0.298** (0.142)	0.271 (0.193)
<i>Panel B: weekly returns</i>						
Additions	1976–2000	434	0.077** (0.037)	0.028*** (0.009)	0.208*** (0.070)	−0.146** (0.070)
	1976–1987	188	0.086* (0.047)	0.026* (0.015)	0.202* (0.113)	−0.162 (0.114)
	1988–2000	246	0.070 (0.055)	0.030*** (0.012)	0.212** (0.087)	−0.134 (0.088)

Table 3 (continued)

Sample		N	Univariate		Bivariate	
			$\Delta\Delta\beta$ (s.e.)	$\Delta\Delta R^2$ (s.e.)	$\Delta\Delta\beta_{SP500}$ (s.e.)	$\Delta\Delta\beta_{nonSP500}$ (s.e.)
Deletions	1979–2000	36	−0.013 (0.157)	−0.030 (0.020)	−0.616** (0.251)	0.771** (0.285)
<i>Panel C: monthly returns</i>						
Additions	1976–1998	300	0.104** (0.047)	0.011 (0.013)	0.173* (0.103)	−0.090 (0.091)
	1976–1987	162	0.054 (0.060)	0.008 (0.020)	0.054 (0.145)	−0.011 (0.125)
	1988–1998	138	0.163** (0.073)	0.015 (0.020)	0.313** (0.144)	−0.183 (0.133)
Deletions	1979–1998	18	0.236 (0.156)	0.057 (0.041)	0.438 (0.315)	−0.133 (0.266)

***, **, and * denote statistical significance at the 1%, 5%, and 10% levels in one-sided tests, respectively.

- The non-S&P coefficient is negative in both groups.

Economic message. Non-synchronous trading contributes something to the daily results, but it cannot be the main story. The comovement shift survives even where the simple non-trading hypothesis predicts it should disappear.

Table 4
Changes in comovement of stocks added to and deleted from the S&P 500 index by change in trading volume

Changes in the slope and the fit of regressions of returns of stocks added to and deleted from the S&P 500 Index on returns of the S&P 500 index and the non-S&P 500 rest of the market. The sample includes stocks added to and deleted from the S&P 500 between 1976 and 2000 that were not involved in mergers or related events (described in the text) and that have sufficient return and trading volume data on CRSP. For each event stock j , the univariate model

$$R_{i,j} = \alpha_j + \beta_j R_{SP500,t} + \epsilon_{i,j}$$

and the bivariate model

$$R_{i,j} = \alpha_j + \beta_{j,SP500} R_{SP500,t} + \beta_{j,nonSP500} R_{nonSP500,t} + \epsilon_{i,j}$$

are separately estimated for the pre-event and post-event period. Returns from October 1987 are excluded. The mechanical influence of the added or deleted stock is removed from the independent variables as appropriate. For the univariate regression model, we examine the mean change in slope across the event date $\Delta\beta$, and the mean change in fit ΔR^2 . For the bivariate model, we examine the mean changes in the slopes, $\Delta\beta_{SP500}$ and $\Delta\beta_{nonSP500}$. The pre- and post-event estimation samples are $[-12, -1]$ and $[1, +12]$ months of daily data. Average pre- and post-event monthly turnover (volume divided by shares outstanding) are computed over these same intervals and are used to identify the direction in trading volume. Panel A reports results for stocks whose turnover decreased. Panel B reports results for stocks whose turnover increased. Standard errors are adjusted using simulations to account for cross-correlation, and are reported in parentheses.

Sample	N	Univariate		Bivariate	
		$\Delta\beta$ (s.e.)	ΔR^2 (s.e.)	$\Delta\beta_{SP500}$ (s.e.)	$\Delta\beta_{non-SP500}$ (s.e.)
Panel A: turnover decrease					
Additions 1976-2000	151	0.083** (0.037)	0.062*** (0.009)	0.113*** (0.044)	-0.374*** (0.055)
1976-1987	65	0.054* (0.036)	0.052*** (0.013)	0.248*** (0.073)	-0.294*** (0.086)
1988-2000	86	0.105** (0.060)	0.070*** (0.012)	0.362*** (0.063)	-0.434*** (0.071)
Panel B: turnover increase					
Additions 1976-2000	295	0.194*** (0.025)	0.045*** (0.007)	0.350*** (0.033)	-0.307*** (0.042)
1976-1987	122	0.089*** (0.031)	0.035*** (0.010)	0.344*** (0.050)	-0.342*** (0.062)
1988-2000	173	0.269*** (0.056)	0.052*** (0.009)	0.354*** (0.044)	-0.282*** (0.056)

***, **, and * denote significant differences from zero at the 1%, 5%, and 10% levels in one-sided tests, respectively.

Table 5
Changes in comovement of stocks added to and deleted from the S&P 500 index: Information diffusion effects (five leads and lags)

Changes in the slope and the fit of regressions of daily returns of stocks added to and deleted from the S&P 500 index on daily returns of the S&P 500 index and the non-S&P 500 rest of the market, using five leads and lags in daily returns to adjust beta for information diffusion effects as suggested by Dimson (1979) and Fowler and Rorke (1983). The sample includes stocks added to and deleted from the S&P 500 between 1976 and 2000 that were not involved in mergers or related events (described in the text) and that have sufficient return data on CRSP. For each event stock j , the univariate model

$$R_{i,j} = \alpha_j + \sum_{k=-5}^5 \beta_j^k R_{SP500,t+k} + \epsilon_{i,j}$$

and the bivariate model

$$R_{i,j} = \alpha_j + \sum_{k=-5}^5 (\beta_{j,SP500}^k R_{SP500,t+k} + \beta_{j,nonSP500}^k R_{nonSP500,t+k}) + \epsilon_{i,j}$$

are separately estimated for the pre-event and post-event period. Returns on the S&P 500 (R_{SP500}) are from the CRSP Index on the S&P 500 Universe file. Returns on a capitalization-weighted index of the non-S&P 500 stocks ($R_{nonSP500}$) in the NYSE, AMEX, and Nasdaq are inferred from the identity described in Table 1. Returns from October 1987 are excluded. The mechanical influence of the added or deleted stock is removed from the independent variables as appropriate. In Panel A, we report the mean difference between the pre-event and post-event Dimson beta (defined as the sum of the lag, contemporaneous, and lead beta coefficients, weighted to account for index autocorrelation as in Fowler and Rorke, 1983). In Panel B, we report the mean difference between the pre-event and post-event components of the Dimson beta (again weighted to account for index autocorrelation). Standard errors are adjusted using simulations to account for cross-correlation, and are reported in parentheses.

Sample	N (Panel A)	Univariate		Bivariate		
		term (Panel B)	$\Delta\beta$ (s.e.)	ΔR^2_{SP500} (s.e.)	$\Delta\beta_{nonSP500}$ (s.e.)	
Panel A: Dimson beta (5 leads and lags)						
Additions 1976-2000	449		0.113*** (0.046)	0.038*** (0.005)	0.110 (0.095)	-0.013 (0.078)
1976-1987	190		0.021 (0.055)	0.028* (0.009)	0.045 (0.116)	0.001 (0.135)
1988-2000	259		0.180*** (0.060)	0.046*** (0.007)	0.158* (0.110)	-0.022 (0.092)
Deletions 1979-2000	76		-0.016 (0.168)	-0.012* (0.008)	-0.627** (0.354)	0.533** (0.255)
Panel B: components of Dimson beta (5 leads and lags)						
Additions 1976-2000	$t = 5$		-0.002 (0.012)	-0.025 (0.001)	0.051*** (0.022)	
	$t = 4$		0.038*** (0.013)	0.059*** (0.025)	-0.039* (0.029)	
	$t = 3$		0.009 (0.012)	-0.002 (0.025)	0.018 (0.027)	
	$t = 2$		0.005 (0.012)	-0.047** (0.023)	0.023 (0.027)	
	$t = 1$		-0.080** (0.014)	-0.118* (0.025)	0.149* (0.031)	
	t		0.150*** (0.021)	0.310*** (0.028)	-0.310*** (0.037)	

Table 5 (continued)

Sample	N (Panel A)	Univariate		Bivariate		
		term (Panel B)	$\Delta\beta$ (s.e.)	ΔR^2 (s.e.)	$\Delta\beta_{SP500}$ (s.e.)	$\Delta\beta_{nonSP500}$ (s.e.)
	$t + 1$		-0.006 (0.023)		-0.070*** (0.013)	0.068** (0.031)
	$t + 2$		0.007 (0.010)		-0.055** (0.024)	0.056** (0.028)
	$t + 3$		-0.021** (0.012)		-0.026* (0.023)	0.048** (0.026)
	$t + 4$		0.023** (0.012)		0.048** (0.024)	-0.025 (0.026)
	$t + 5$		0.000 (0.012)		0.044* (0.023)	-0.051 (0.025)

***, **, and * denote significant differences from zero at the 1%, 5%, and 10% levels in one-sided tests, respectively.

5.5 Table 5: information diffusion and Dimson betas

Read:

- After adjusting for five leads and lags, the daily univariate beta shift for additions falls from about **0.151** to about **0.113**.
- The daily bivariate S&P beta shift falls much more sharply, from about **0.326** to about **0.110**.
- The adjusted non-S&P beta shift becomes small and statistically weak in the full addition sample.
- The lead-lag decomposition shows that much of the information-diffusion effect is concentrated around **one- and two-day lags**, not long horizons.

Economic message. Information diffusion explains a meaningful share of the **daily** effect, especially in the bivariate regressions, but a substantial residual remains. That residual is the evidence most naturally attributed to category and habitat demand.

6 Short summary

- The paper asks whether comovement is driven only by common fundamentals or also by market structure and investor demand.
- S&P 500 additions provide a natural reclassification experiment.

- After inclusion, stocks load more on S&P returns and less on non-S&P returns.
- The effect is stronger in more recent years, when S&P-linked investing is more important.
- Matching-stock tests, calendar-time tests, and non-trading checks leave the main result intact.
- Lead-lag tests show that information diffusion matters, but it does not explain everything.
- The paper’s bottom line is that index membership can change a stock’s beta because investors trade the index as a category or habitat.

7 Conclusion

7.1 Core conclusion

The paper rejects a purely fundamentals-based account of return comovement. Stocks added to the S&P 500 start moving more with S&P stocks and less with non-S&P stocks, and this happens in a way that is difficult to reconcile with unchanged cash-flow covariance.

7.2 Economic intuition

The S&P 500 is more than a statistical basket of firms. It is a focal trading object. Once a stock becomes part of that object, it is exposed to the fund flows, trading clienteles, and information environment attached to the index. Those forces reshape its covariance with other stocks.

7.3 Takeaway

This paper demonstrates that betas are not purely “fundamental” objects. Market institutions, index construction, and investor behavior can change how assets move together.